SHREEYAN RIJAL

Computational Materials Physicist | Prospective PhD Researcher — Photovoltaic & Energy Materials

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Research Profile

Final-year B.Sc. Physics student with three first-author peer-reviewed publications in computational materials science. Applies first-principles (DFT) methods to perovskite photovoltaics and van der Waals semiconductor heterostructures, with emphasis on optoelectronic and thermoelectric properties relevant to energy-conversion materials. Seeking a fully funded PhD position to develop computational and device-modeling approaches for next-generation solar cell and energy materials.

Education

B.Sc. Physics — St. Xavier's College, Tribhuvan University, Kathmandu, Nepal **2021–2025**

Coursework: Quantum Mechanics, Condensed Matter Physics, Electrodynamics, Thermodynamics, Computational Methods

Research Experience

Independent Researcher, Computational Materials Physics — St. Xavier's College **2022–Present**

- Performed DFT modeling (Quantum ESPRESSO) and device-level simulation (OghmaNano) of perovskite and thin-film solar cell architectures under thermal and optical stress, identifying parameter regimes that improve stability and efficiency.
- Carried out spin-polarized first-principles calculations of Mn-doped silicon, characterizing impurity-induced near-metallicity and enhanced infrared optical activity relevant to spintronic device design.
- Modeled strain-engineered optoelectronic and thermoelectric properties of van der Waals heterostructures (MoSSe/Ti2CO2, Hf2CO2/blue phosphorus, Zr2CO2/blue phosphorus) using first-principles methods.
- Collaborated with multi-institutional research teams, producing three first-author peer-reviewed publications and several manuscripts in preparation.

Publications (Peer-Reviewed, First Author)

1. **Rijal, S.**, Sah, N., Chaudhary, P., & Jha, V.K. (2026). Enhancing the stability and efficiency of perovskite solar cells by optimization of electrical parameter under thermal and optical stress. *Optical and Quantum Electronics* (Springer), 58, 288. DOI: 10.1007/s11082-026-08876-3
2. **Rijal, S.**, Jha, V.K., et al. (2026). Impurity-induced near-metallicity, spin-polarized transport, and enhanced infrared optical activity in Mn-doped silicon: A spin-polarized first-principles study. *Physica B: Condensed Matter*, 739, 418967. DOI: 10.1016/j.physb.2026.418967
3. **Rijal, S.**, Chaudhary, P., Sharma, S., Thada Magar, M.T., Regmi, B., Adhikari, B. et al. (2026). Time–frequency characterization of prompt penetration electric fields across distinct geomagnetic regimes. *Advances in Space Research* (In Press). DOI: 10.1016/j.asr.2026.06.060

Selected Ongoing Projects

- Strain-Engineered Optoelectronic and Thermoelectric Properties of MoSSe/Ti2CO2 van der Waals Heterostructure: A First-Principles Study — Journal of Physics and Chemistry of Solids (Under Review, Lead Author)
- Strain-Engineered Optoelectronic and Thermoelectric Properties of Hf_2CO_2/Blue Phosphorus and Zr_2CO_2/Blue Phosphorus van der Waals Heterostructures: First-Principles Studies — In Progress, Lead Author (two manuscripts)
- Double Perovskite Energy Material Study — Advanced Energy Materials (Under Review, Co-Author)

Technical Skills

DFT & Simulation: Quantum ESPRESSO, Gaussian, OghmaNano (device simulation), ORCA (collaborator)
Programming & Analysis: Python (data analysis, machine learning, automation), MATLAB, C, LaTeX

Physics Areas: Condensed matter and semiconductor physics, spintronics, photovoltaics, optoelectronics, thermoelectrics

Leadership & Mentorship

- Research Mentor, St. Xavier's College (2023–Present) — Guide junior undergraduates into DFT research and scientific writing.
- LSRA Mentor (2024–Present) — Mentor a candidate for the Fathers Locke & Stiller Research Award.
- Library Council Representative, Physics Department, St. Xavier's College (2023–Present)

Languages

Nepali (Native) • English (Fluent, Professional Proficiency) • Hindi (Fluent, Professional Proficiency)

References

Dr. Vinaya Kumar Jha — Research Supervisor, St. Xavier's College, Tribhuvan University
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Dr. Binod Adhikari — Research Collaborator
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Dr. Basu Dev Ghimire — Research Collaborator
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